

Le système international d'unités The International System of Units

9 édition 2019

Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du mètre
et aux représentations secondaires de la seconde

**Recommended values of standard frequencies for
applications including the practical realization of the
metre and secondary representations of the second**

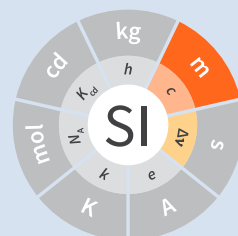
Partie 2
Part 2

Helium Neon Laser (unstabilized) ()

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CCL-CCTF Frequency Standards Working Group

August 2012
août 2012



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Helium Neon Laser (unstabilized) ()**

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1. CIPM recommended value [1] of the frequency

$$f(\text{HeNe}_{\text{unstabilised}}) = 473.6127 \text{ THz}$$

equivalent to

$$\lambda(\text{HeNe}_{\text{unstabilised}}) = 632.9908 \text{ nm}$$

with a relative standard uncertainty of 1.5×10^{-6} applies apply to the radiation in vacuum of an unstabilized helium-neon laser operating solely on the $3s_2 \rightarrow 2p_4$ transition, independent of the isotopic mixture of the neon.

This wavelength (in vacuum) value was also recommended by CIPM 2007 as a Realization of the Definition of the Metre

2. Source data

The source data are derived from an investigation carried out by the CCL and published in [2]

HeNe laser operating on the $3s_2 \rightarrow 2p_4$ transition

References

- [1] Procès-Verbaux des Séances du Comité International des Poids et Mesures, 96th meeting (2007) 2009, Recommendation 2 (CI-2007): On the value and uncertainty of unstabilized He-Ne lasers, page 186. (see e.g. <http://www.bipm.org/utls/en/pdf/CIPM2007-EN.pdf#page=78>).
- [2] J. A. Stone, J. E. Decker, P. Gill, P. Juncar, A. Lewis, G. D. Rovera, M. Viliésid: Advice from the CCL on the use of unstabilized lasers as standards of wavelength: the helium–neon laser at 633 nm. *Metrologia* **46**, 11–18 (2009).